Ballot Manifest Verification

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Abstract

Public vote totals are not meaningful unless the counted inputs are accounted for. This paper defines the first RCOUNT ballot-manifest verification layer: batch manifest records must reconcile accepted, counted, and rejected ballot counts, and each batch-level summary must reference an existing manifest row. Using the `mail-batch-added` and `missing-batch` fixtures, we show how RCOUNT can distinguish a legitimate late mail batch from a canvassed summary that points at absent batch evidence.

1 Introduction

Election totals are often published as precinct or jurisdiction summaries, but the operational evidence underneath them is batch-based. Election-day scanner batches, central-count mail batches, provisional batches, duplicated-ballot batches, and vote-center batches may all contribute to the same contest total. If the public summary says a batch was counted, the package needs a manifest row for that batch.

V.5 defines the first RCOUNT ballot-manifest verification layer. It does not try to prove voter eligibility or voter intent. It asks a narrower accounting question: do the batch manifest rows and batch-level summaries reconcile?

This makes the ballot manifest an accounting control. It is not the same object as a voter list, ballot image, ballot card, cast-vote record, contest record, or candidate selection. RCOUNT keeps those categories separate so a verifier can say exactly which layer passed and which layer still needs evidence.

2 Manifest Model

The current RCOUNT batch manifest row is intentionally small:

Field	Meaning
`batch_id`	Stable package identifier for the batch.
`reporting_unit_id`	Public unit associated with the batch, such as a precinct or central-count unit.
`kind`	Operational class, such as election-day, mail, provisional, vote-center, or central-count.
`status`	Publication status represented in the package.
`accepted_ballots`	Ballots accepted into the batch's accounting path.
`counted_ballots`	Ballots counted from that batch for the represented contest summaries.
`rejected_ballots`	Accepted-accounting records later rejected or removed from counting.
`source_refs`	Source evidence rows that support the manifest row.

Object	Boundary
Voter	Eligibility and participation records are not public batch manifest rows.
Ballot or ballot card	Physical or electronic voting artifacts may be represented by manifests, but are not identical to batch rows.
CVR row	A cast-vote record records contest interpretation; V.6 reconciles CVRs to summaries.
Contest summary	A contest summary can reference a batch, but it is not the manifest itself.
Selection or vote	Candidate choices, write-ins, overvotes, undervotes, and blanks are contest semantics, not batch identity.

The package contract is:

- normalized batch manifest rows live at `normalized/batches.ndjson`;
- summaries that claim batch-level support carry a `batch_id`;
- `batch_id` is stable inside the package and may be mapped from vendor ids, scanner labels, or election-office labels by an adapter;
- `source_refs` bind the normalized row back to hashed source evidence in `sources/source-index.json`.

Vendor export adapters and external schema mapping belong to V.9. V.5 only defines the normalized RCOUNT contract those adapters must satisfy.

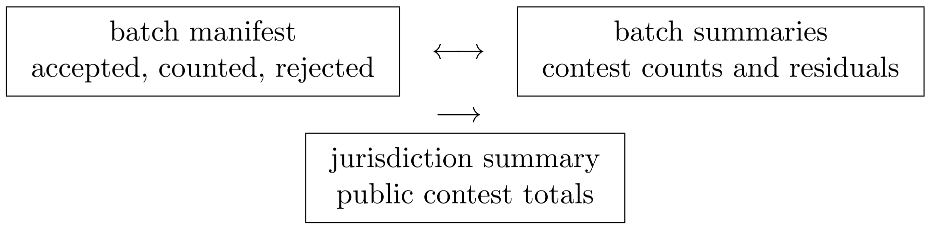
3 Verification Equations

RCOUNT currently exposes two manifest-layer equations:

$$\texttt{accepted\_ballots(batch)} = \texttt{counted\_ballots(batch)} + \texttt{rejected\_ballots(batch)}$$

$$\texttt{batch\_summary\_total(batch)} = \texttt{counted\_ballots(batch)}$$

The second equation is only evaluated for summary rows with a `batch_id`. The verifier first checks that the batch id exists in `normalized/batches.ndjson`. If a summary references a missing batch, the package fails before the batch total can be trusted.



Jurisdiction totals roll up only after these local controls hold. A verifier that silently treats a missing batch as zero, optional, or inferred would hide the exact error the manifest layer is designed to catch.

4 Fixtures

The positive fixture is `mail-batch-added`. It represents a canvassed state where precinct P-001 has an election-day batch and a late mail batch. The status event explains that the late mail batch was accepted before canvass.

Batch	Accepted	Counted	Rejected
`batch:P-001:election-day`	70	70	0
`batch:P-001:late-mail`	10	10	0
`batch:P-002:election-day`	60	60	0

The fixture uses these anchors:

Path	Role
`normalized/batches.ndjson`	Declares batch manifest rows.
`normalized/summaries.ndjson`	Declares batch-level and jurisdiction summaries.
`status/events.ndjson`	Explains the late-mail transition into the canvassed state.
`transcripts/verify-transcript.json`	Records verifier pass/fail checks.

The negative fixture is `missing-batch`. Its summaries still reference the P-001 late-mail batch, but `normalized/batches.ndjson` omits that manifest row. Source hashes and package structure can still be present; the accounting evidence is incomplete.

5 Verifier Traceability

The fixtures can be regenerated and checked directly:

```
cargo run -p rcount-io --example mail_batch_added_package
cargo run -p rcount-io --example missing_batch_package
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/mail-batch-added
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/missing-batch
```

Fixture	Expected result	Carrying checks
`mail-batch-added`	transcript status `pass`	`accepted_ballots` and `batch_summary_total` pass for each declared batch.
`missing-batch`	transcript status `fail`	`batch_summary_total` fails because the P-001 late mail summary references an absent batch.

The positive transcript also preserves the larger audit context: jurisdiction total and source-hash checks pass after batch accounting passes. The negative transcript shows the opposite shape: source and package evidence can be present while the batch accounting check still fails.

The negative transcript says:


```
{
  "equation_id": "batch_summary_total",
  "status": "fail",
  "error": "summary for contest syn-2024-mayor reporting unit \
syn:precinct:P-001 references missing batch id: batch:P-001:late-mail"
}
```


6 Boundaries

Ballot manifest verification does not certify an election. It verifies a public accounting relationship inside the package. Officials, canvassing boards, recount procedures, and courts still decide eligibility, acceptance, cure, duplication, adjudication, recount changes, and certification.

Workflow event	Manifest implication
Late mail accepted	Add or update a mail batch with accepted and counted ballots.
Provisional review	Add, reject, or reclassify provisional batch evidence after eligibility decisions.
Ballot cure	Move records into accepted accounting only after the cure decision.
Duplicated ballot resolution	Link counted batch evidence to duplication/adjudication source records.
Recount or court order	Publish amended batch and summary records with event lineage.

A missing batch is therefore a verification failure, not a standalone fraud finding. Investigation determines whether the cause is an export defect, publication lag, data-entry omission, lawful amendment not yet represented, or misconduct.

The manifest layer also does not prove voter intent or tabulator correctness. A batch may reconcile numerically while a later CVR or audit layer still finds a parser, contest-definition, or hand-audit problem. RCOUNT should report those as separate failures instead of collapsing them into one generic claim.

Finally, batch granularity has privacy implications. Very small batches, single-voter provisional batches, rare write-in batches, or precise batch times can create small-cell disclosure risk. Publication policy must be reviewed with V.4's receipt-safety rule in mind.

Publication pattern	Privacy risk
Single-voter provisional batch	Can reveal whether one person's ballot was accepted or rejected.
Tiny cure batch	Can expose timing and status of a small group of voters.
Rare write-in or contest batch	Can narrow candidate-choice inference in a small cell.
Precise scanner time plus small unit	Can combine with observation or check-in records.

Jurisdictions may need public/restricted manifest views, aggregation, redaction, or delayed publication. Hashes and source references provide tamper evidence; they do not by themselves provide privacy.

7 Conclusion

Ballot manifest verification is the accounting floor under public totals. A late mail batch can be legitimate when it is represented by manifest evidence, status events, source references, and reconciled summaries. A summary that points at a missing batch is a mechanical failure.

That distinction is what RCOUNT is for: not to replace canvass judgment, but to make the public evidence behind canvass totals replayable and falsifiable. V.6 can now move inside the counted batches to reconcile cast-vote records to summaries, and V.7 can use the manifest as one input to replayable risk-limiting audit workflows.

References

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